

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<thought>

To solve this problem , we need to understand what CD 2 1 does and how it might impact the immune response , particularly against viruses . CD 2 1 is an important molecule in B cell activation and the complement system 's regulation . It is involved in the attachment of HIV to B cells .

</thought>

< answer >

C . Human immunodeficiency virus (HIV)</answer>

<thought>

HIV can attach to B cells by using CD 2 1 on the cell surface . The problem states that the scientist genetically modifies B cells so that they do not express CD 2 1 . This modification would prevent HIV from effectively attaching to and infecting B cells . Hence , this genetic modification would most likely affect the pathogenesis of HIV .

</thought>

